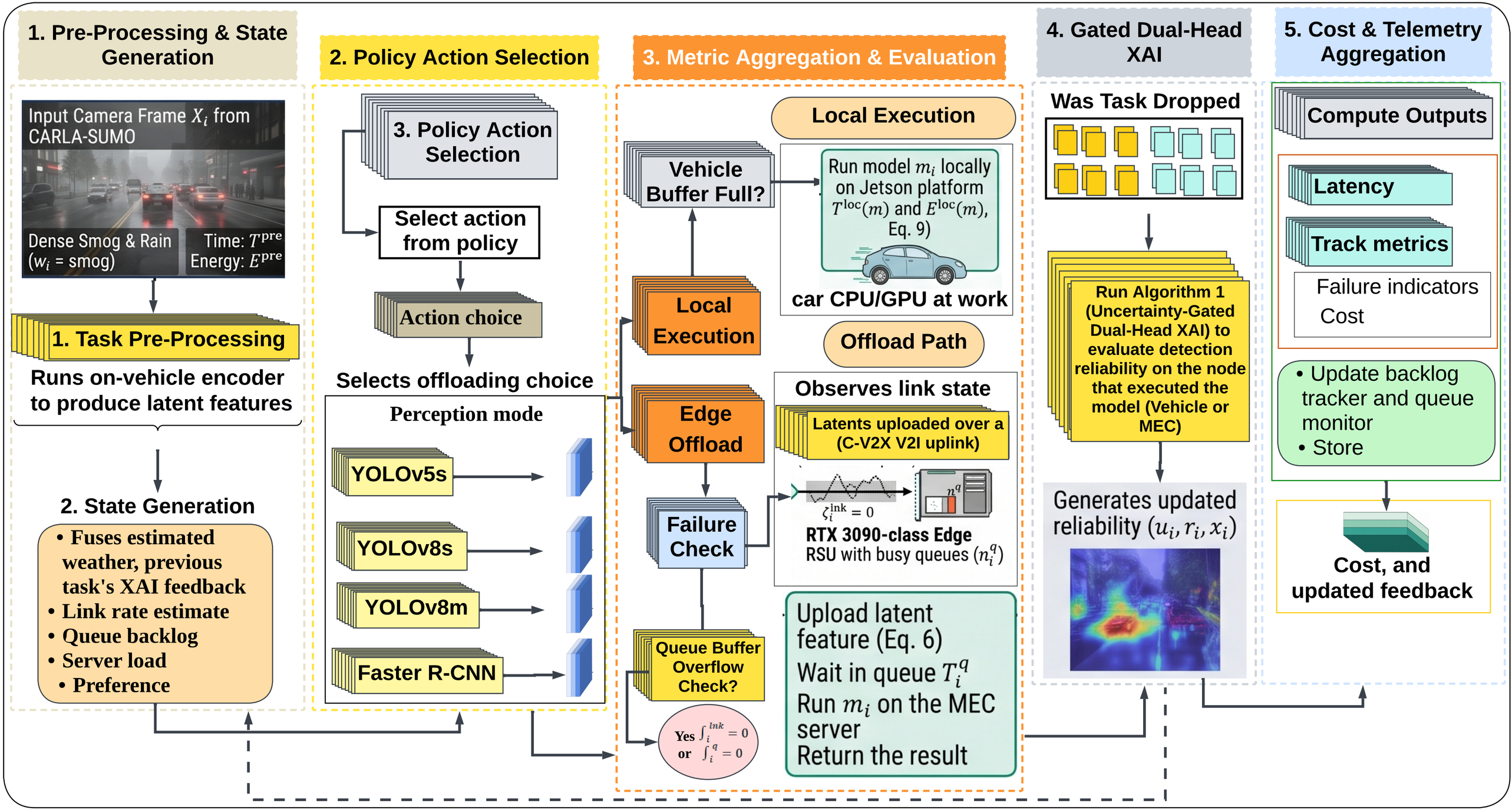

1. Pre-Processing & State Generation
2. Policy Action Selection
3. Metric Aggregation & Evaluation
4. Gated Dual-Head XAI
5. Cost & Telemetry Aggregation
Input Camera Frame X_i from CARLA-SUMO
Dense Smog & Rain (w_i = smog)
Time: T^{pre}
Energy: E^{pre}
1. Task Pre-Processing
Runs on-vehicle encoder to produce latent features
2. State Generation
• Fuses estimated weather, previous task's XAI feedback
• Link rate estimate
• Queue backlog
• Server load
• Preference
3. Policy Action Selection
Select action from policy
Action choice
Selects offloading choice
Perception mode
YOLOv5s
YOLOv8s
YOLOv8m
Faster R-CNN
Local Execution
Vehicle Buffer Full?
Run model m_i locally on Jetson platform $T^{loc}(m)$ and $E^{loc}(m)$, Eq. 9)
car CPU/GPU at work
Local Execution
Offload Path
Edge Offload
Observes link state
Latents uploaded over a (C-V2X V2I uplink)
$\zeta_i^{ink} = 0$
n^q
RTX 3090-class Edge
RSU with busy queues (n_i^q)
Failure Check
Queue Buffer Overflow Check?
Yes $\int_i^{lnk} = 0$ or $\int_i^q = 0$
Upload latent feature (Eq. 6)
Wait in queue T_i^q
Run m_i on the MEC server
Return the result
Was Task Dropped
Run Algorithm 1 (Uncertainty-Gated Dual-Head XAI) to evaluate detection reliability on the node that executed the model (Vehicle or MEC)
Generates updated reliability (u_i, r_i, x_i)
Compute Outputs
Latency
Track metrics
Failure indicators
Cost
• Update backlog tracker and queue monitor
• Store
Cost, and updated feedback